

Regular Expressions

+ Digit = 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

+ Unsigned = Digit Digit*

+ Integer = (- | ε) Unsigned

+ Real = Integer (ε | . Integer)

+ Num = Real | Integer

+ Letter = A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S
| T | U | V | W | X | Y | Z | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o
| p | q | r | s | t | u | v | w | x | y | z

+ ID = Letter (Letter | Digit)*

Tokens

==	!=	=	<=	>=
<	>	;	,	(
)	[	]	{	}
+	-	*	/	void
real	int	return	if	else
while	Num	ID		

Grammar Rules

- + program $\rightarrow$ declaration-list
- + declaration-list $\rightarrow$ declaration declaration-list | declaration
- + declaration $\rightarrow$ var-declaration | fun-declaration
- + var-declaration $\rightarrow$ type-specifier ID ; | type-specifier ID [Num] ;
- + type-specifier $\rightarrow$ int | real | void
- + fun-declaration $\rightarrow$ ID type-specifier (params) compound-stmt
- + params $\rightarrow$ param-list | ϵ
- + param-list $\rightarrow$ param param-list | param
- + param $\rightarrow$ type-specifier ID | type-specifier ID []
- + compound-stmt $\rightarrow$ { local-declarations stmt-list }
- + local-declarations $\rightarrow$ var-declaration local-declarations | ϵ
- + stmt-list $\rightarrow$ statement stmt-list | ϵ
- + statement $\rightarrow$ expression-statement | compound-statement |
selection statement | iteration-statement | return statement
- + expression-stmt $\rightarrow$ expression ; | ;
- + selection-statement $\rightarrow$ if (expression) statement | if (expression)
statement else statement
- + iteration-stmt $\rightarrow$ while (expression) statement

- + return-stmt $\rightarrow$ return ; | return expression ;
- + expression $\rightarrow$ var = expression | simple-expression
- + var $\rightarrow$ ID | ID [expression]
- + simple-expression $\rightarrow$ additive-expression relOp additive-expression |
additive-expression
- + relOp $\rightarrow$ <= | >= | < | > | != | ==
- + additive-expression $\rightarrow$ term addOp additive-expression | term
- + addOp $\rightarrow$ + | -
- + term $\rightarrow$ factor mulOp term | factor
- + mulOp $\rightarrow$ * | \
- + factor $\rightarrow$ (expression) | var | call | Num
- + call $\rightarrow$ ID(args)
- + args $\rightarrow$ args-list | ϵ
- + args-list $\rightarrow$ expression , args-list | expression